MILITARY SPECIFICATION

PROJECTILE, HEP-T, 105MM, M393A2 METAL PARTS

1. SCOPE

1.1 This specification covers Projectile, HEP-T, 105MM, M393A2, Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-application

STANDARDS

Federal

FED-STD-141	- Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing
FED-STD-151	- Metal, Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Qualification of Inspection Personnel (Magnetic Particle and Penetrant)

FSC 1315

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2.1 (Cont'd)

MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F8837835	- Pallet
F10524305	- Projectile, HEP-T, 105MM, M393A2, Metal Parts
IEL-10524305	- Inspection Equipment List for Projectile, HEP-T, 105MM, M393A2, Metal Parts

PUBLICATIONS

Frankford Arsenal

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection, including Drawing D8593429.

Picatinny Arsenal

PA-PD-792	- Equipment, Testing or Measuring for Ammunition Items

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

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3.2 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.2.1 Hot forge heat treat method.-Prior to final machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees fahrenheit (F), followed by oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees (F), suitable to produce the required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees (F), afterwhich cooling may be accelerated by water spray.

3.2.2 Cold shaping and hot cup - cold draw method.-Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700 degrees (F), minimum for a minimum of 30 minutes following the final forming operation during which time plastic flow takes place.

3.3 Mechanical properties.

3.3.1 Body and base plug.-Mechanical properties specified by the applicable drawings shall apply throughout each component.

3.3.2 Rotating bands.-After application and subsequent removal from the projectile, the rotating bands shall have the following mechanical properties:

Tensile strength 32,000 pounds per square inch (psi) min.

Elongation 15 percent per inch of gage length.

3.4 Banding.

3.4.1 Band seat. -Prior to assembly of rotating band blanks, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seats after annealing.

3.4.2 Band blanks.-The rotating band blanks shall be in the form of a solid ring, annealed if necessary to comply with requirements of the applicable specification.

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3.4.3 Application.-The rotating band blanks shall be applied after final heat treatment of the body by carefully pressing them into the band seats so that contact is made with the sidewalls for a minimum of eight-tenths of the band seat depth. There shall be no gap between band and sidewall at the outside surfaces of the body after final machining of rotating band.

3.4.4 Band seating.-The diametrical clearance between the rotating band and band seat shall not exceed .006 inch (See 6.3).

3.5 Metal defects.

3.5.1 Body.-The body, including the cavity, shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, bursts folds, seams and other metal defects.

3.5.2 Body cavity.- The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.5.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first month's production. However, the establishment of visual standards will not waive the requirement for compliance with 3.5.1 and 3.5.2 (See 6.2)

3.6 Proving ground functional test.-The projectile assembly shall reveal no evidence of failure, when tested at excess and service pressure. Metal parts failure shall be:

a. Permanent deformation greater than .010 inches on either the body diameter or bourrelet diameter of the projectile.

b. Loss or break-up of metal parts components.

c. Loose or fanning of rotating band(s).

d. Shear of rotating band(s) as evidenced by complete lack of engraving.

e. Leakage of propellant gases into the base plug thread area.

3.7 Protective coating.

3.7.1 Base plug.-All aspects of surface pretreatment of surface coating are as shown by the applicable specification, subject to demonstration of control of processes thru salt spray test of panels. Requirements of the component drawing apply for period of salt spray exposure without coating breakdown.

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3.7.2 Body assembly.-Requirements relating to paint thickness are not specified, except as a guide for performance of tests as indicated by the applicable drawing and specifications.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.8.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.8.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.-All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

3.8.4 Rotating band damage.-The rotating band after machining shall be protected from being nicked, burred or otherwise damaged during subsequent handling and preparation for shipment.

3.8.5 Compatibility.-Corrosion preventive compound specified on the applicable drawing shall be compatible with the high explosive filler to be used. Information relative to compatibility can be obtained from the responsible technical agency by request through the procuring activity. See 6.4 for approved sources.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

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4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground functional test. (See 3.6).

b. Acceptance or rejection of lot by the Government shall be contingent on successful passage of the proving ground function tests in 4.3.4.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 35 metal parts assemblies as follows:

25 metal parts assemblies complete (painted)
5 metal parts assemblies complete (unpainted) including base plugs
5 sets of component parts unassembled and unpainted including rotating band blanks (bodies to be completely machined, including band seat)

Twenty of the above painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

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4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain bodies forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Stress relieved lots.-Stress relieved lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Miscellaneous lots.-Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.1.2 or 4.3.1.3, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.5 Projectile lots.-When it has been determined that bodies of a heat-treated, stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or workweek. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235

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4.3.2 (Cont'd)

may be used if approved by the procuring activity. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual inspection for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Control tool manufacture.-Dimensions marked tool control shall be gaged at the beginning of each eight hours production or fraction thereof from each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.3.2.2 Plug, Base, Drawing D10524308, covering a detail of Drawing F10524305.

Defect	Method of Inspection
CRITICAL	
Metal defective	Visual
MAJOR	
101. Concentricity of pitch diameter of large internal thread, with pitch diameter of external thread, including perpendicularity of front face of shoulder and bottom of large internal thread cavity	Gage
102. Concentricity of large outside diameter with pitch diameter of external thread, including perpendicularity of front face of shoulder	Gage
103. Concentricity of pitch diameter of small internal thread with pitch diameter of external thread, including perpendicularity of front face of shoulder and rear face	Gage
104. Perpendicularity of front face of shoulder with pitch diameter of external thread	Gage
105. Flatness of front face of shoulder	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
106.	Perpendicularity of rear face with pitch diameter of small internal thread	Gage
107.	Pitch diameter of external thread, min.	Gage
108.	Major diameter of external thread, min.	Gage
109.	Pitch diameter of large internal thread, max.	Gage
110.	Minor diameter of large internal thread, max.	Gage
111.	Pitch diameter of small internal thread, max.	Gage
112.	Minor diameter of small internal thread, max.	Gage
113.	Pitch diameter of external thread, max.	Gage
114.	Pitch diameter of large internal thread, min.	Gage
115.	Pitch diameter of small internal thread, min.	Gage
116.	Length of effective small internal thread, min.	Gage
117.	Depth from front of shoulder to bottom of large cavity	Gage
118.	Length of ineffective external thread, max. (if applicable)	Gage
119.	Length of ineffective large internal thread, max. (if applicable)	Gage
120.	Depth of tracer cavity counterbore, max.	Gage
121.	Thickness at juncture of cavity and sidewall, min.	Gage
122.	Diameter of counterbore in large internal cavity	Gage
123.	Depth of counterbore in large cavity, max.	Gage
124.	Thread(s) damaged or not full	Visual
MINOR		
201.	Diameter of shoulder	Gage
202.	Diameter at rear of taper	Gage
203.	Length from front face to shoulder	Gage
204.	Length, total	Gage
205.	Thickness of shoulder	Gage
206.	Presence and location of wrench holes	Visual (See 6.2)
207.	Radii or chamfers missing or incorrect	Visual (See 6.2)
208.	Surface finish improper	Visual (See 6.2)
209.	Protective coating inadequate or damaged	Visual
210.	Burr	Visual
211.	Foreign matter	Visual

4.3.2.3 Body, Part No. 10524307 covering a detail of Drawing F10524305

CRITICAL		
	Metal defective	Visual (Probe light)

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of forward rotating band seat	Gage
102. Diameter of rear rotating band seat	Gage
103. Width of forward rotating band seat	Gage
104. Width of rear rotating band seat	Gage
MINOR	
201. Location of forward rotating band seat	Gage
202. Location of rear rotating band seat	Gage
203. Scoring missing from rotating band seat ribs	Visual
204. Foreign matter on rotating band seat	Visual

4.3.2.4 Body assembly, Drawing F10524306, covering a detail of Drawing F10524305.

CRITICAL	
Metal Defective (After Phosphating)	Visual (Probe light)
MAJOR	
101. Concentricity of ogive with bourrelets	Gage
102. Concentricity of cavity forward of base plug thread with bourrelets	Gage
103. Concentricity of forward rotating band with bourrelets	Gage
104. Concentricity of rear rotating band with bourrelets	Gage
105. Concentricity of pitch diameter of base plug thread with bourrelets, including perpendicularity of rear face	Gage
106. Small counterbore not functionally concentric with large counterbore	Gage
107. Large counterbore not functionally concentric with pitch diameter of thread	Gage
108. Flatness of rear face	Gage
109. Perpendicularity of rear face with pitch diameter of base plug thread	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
110.	Diameter of bourrelets, min.	Gage
111.	Diameter of cavity forward of base plug thread	Gage
112.	Diameter of body midsection, min. (for indicated length)	Gage
113.	Diameter forward of front rotating band, min. (for indicated length)	Gage
114.	Diameter between rotating bands, min.	Gage
115.	Diameter at rear of rear rotating band, min.	Gage
116.	Diameter of forward rotating band, flat	Gage
117.	Diameter of rear rotating band, flat	Gage
118.	Depth of large counterbore	Gage
119.	Depth of effective thread, including large and small counterbore	Gage
120.	Location of forward rotating band	Gage
121.	Location of rear rotating band	Gage
122.	Profile of nose ogive	Gage
123.	Thickness of wall at front, min.	Gage
124.	Thickness of wall in ogive area	Gage
125.	Thickness of wall at forward datum location, min.	Gage
126.	Thickness of wall at center datum location	Gage
127.	Thickness of wall at rear datum location	Gage
128.	Variation in wall thickness in ogive area excessive	Gage
129.	Variation in wall thickness at forward datum location excessive	Gage
130.	Foreign matter in cavity	Visual
MINOR		
201.	Diameter of large counterbore	Gage
202.	Diameter of small counterbore	Gage
203.	Depth from rear face to bottom of small counterbore	Gage
204.	Length of ineffective thread, max. including large and small counterbores	Gage
205.	Width of forward rotating band	Gage
206.	Width of rear rotating band	Gage
207.	Length, total	Gage
208.	Depth of undercut front and rear of forward rotating band (if applicable)	Visual (See 6.1)

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
209.	Depth of undercut front and rear of rear rotating band (if applicable)	Visual (see 6.1)
210.	Width of undercut front and rear of forward rotating band (if applicable)	Visual (see 6.1)
211.	Width of undercut front and rear of rear rotating band (if applicable)	Visual (see 6.1)
212.	Radius of nose not properly blended	Visual (see 6.1)
213.	Radii or chamfers missing or incorrect	Visual (see 6.1)
214.	Surface finish improper	Visual (see 6.1)
215.	Burr	Visual
216.	Foreign matter (unless otherwise classified)	Visual

4.3.2.5 Metal Parts (after painting), Drawing F10524305.

CRITICAL

1.	Hydrostatic pressure stamp missing (signifying failure to pass hydrostatic test)	Visual
2.	Metal defective	Visual (probe light)

MAJOR

101.	Weight	Approved scale
102.	Pitch diameter of thread, max.	Gage
103.	Pitch diameter of thread, min.	Gage
104.	Minor diameter of thread, max.	Gage
105.	Diameter forward of rotating bands, max.	Gage
106.	Diameter rear of rotating bands, max.	Gage
107.	Paint in cavity inadequate or excessive	Visual

MINOR

201.	Protective coating inadequate or damaged	Visual
202.	Paint on thread(s)	Visual
203.	Paint on undercut(s)	Visual
204.	Paint on rear face	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
205. Paint on rotating bands bearing surface(s)	Visual
206. Rotating band(s) damaged	Visual
207. Silicone compound missing from indicated areas	Visual
208. Marking missing, incorrect or illegible	Visual
209. Foreign matter	Visual
210. Protector cap missing	Visual

4.3.2.6 Carton, packed (base plug) prior to sealing.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Support missing or damaged	Visual
202. Separator(s) missing or damaged	Visual
203. Improperly packed	Visual

After Sealing

MINOR

201. Contents loose	Manual
202. Carton damaged, cut or gouged thru all layers	Visual
203. Carton not properly sealed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.2.7 Pallet.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Strapping missing, loose or broken	Manual/Visual
202. Pallet damaged	Visual
203. Improperly packed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a rockwell hardness test or any other method of hardness testing approved by the procuring activity.

4.3.3.1.1.2 Stress relieved lots.-Twenty-five cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.3.1.3) processed through the stress relief furnace and 5 bodies from each one hours production thereafter for the balance of the stress relieved lot shall be subjected to a rockwell hardness test, or any other method of hardness testing approved by the procuring activity to assess uniformity of the product.

4.3.3.1.1.3 Miscellaneous lots.-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.3.1.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.3.3.1.1.2.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall

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4.3.3.1.2.1 (Cont'd)

be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.-Two projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five degrees fahrenheit but not below the minimum temperature of 700 degrees fahrenheit and the minimum 30 minutes soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each 8 hours by an independent method.

4.3.3.1.2.3 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.3.3.2.1 or 4.3.3.2.2 as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.3.1.2.4 Rotating bands.-One body assembly shall be selected for each 8 hours production or fraction thereof, from each banding machine in operation and subjected to the tension test. Failure of either band to comply with 3.3.2 shall be cause for rejection of the production represented by the sample.

4.3.3.1.2.5 Base plug.-Two base plugs from each 1000 base plugs or fraction thereof shall be subjected to this test to determine compliance with the mechanical properties requirement of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot of components represented by the failing specimen. At the discretion of the procuring activity this test may be performed on the raw material, using one test specimen for each 1000 pounds of material or fraction thereof, except that the test performed by the Government shall be made on specimens taken from the finished components.

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4.3.3.2 Hydrostatic Test (a critical defect).-After cleaning but prior to surface treatment and painting, each metal parts assembly shall be subjected to a hydrostatic pressure to determine compliance with the requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection. Assemblies which comply shall be stamped "H" to signify compliance.

4.3.3.2.1 Profile of ogive.-After successfully passing the hydrostatic test, the sample supplied in accordance with 4.3.2.4 shall be gaged with an approved profile gage. Failure of any projectile to comply with the requirements of the drawing shall be cause for rejection of the lot represented by the sample.

4.3.3.3 Magnetic particle inspection.

4.3.3.3.1 Heat treated or stress relief bodies after final machining prior to banding, shall be inspected by magnetic particle inspection techniques approved by the responsible technical.

a. 100 percent magnetic particle inspection will be in effect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

b. Reduced inspection may then be instituted for magnetic particle inspection consisting of 60 bodies per hour for ten consecutive working days.

c. If no defectives are found during the ten days, the sample may be reduced to 40 bodies per hour for 5 consecutive working days.

d. If no defects are encountered during the five days, the sample may be reduced to 20 bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) the magnetic particle inspection shall revert back to (a) above, in addition a 100 percent back screening to the last acceptable hours production will be performed. If one or more defects are discovered in the back screening of the last acceptable hours production, continued back screening will be in affect until an acceptable hours production is reached,i.e., no defects discovered in screening operations.

f. For requalification for reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a) through (d) above.

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g. Should a controversy arise as to the validity of an indication, the contractor shall use metallographic test methods or an approved non-destructive method for proof of acceptability, to the responsible technical agency.

h. Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as a alternate procedure in lieu of corrective manufacturing process action to eliminate a continuing process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until the contractor can prove his process is capable of producing suspect free projectiles.

4.3.3.4 Band seating.

4.3.3.4.1 Tightness.

4.3.3.4.1.1 Non-destructive method.-Ten body assemblies from each one hour's production per banding machine in operation shall be subjected to this test. Failure of any band to comply with 3.4.4 shall be cause for rejection of the assemblies represented by the sample.

4.3.3.4.1.2 Destructive method.-When the nondestructive equipment is not available, two body assemblies shall be selected from each 8 hour production or fraction thereof per banding machine in operation. Failure of any band to comply with 3.4.4 shall be cause for rejection of the production represented by the sample (see 6.3).

4.3.3.4.2 Sidewall contact.-Rotating bands as specified in paragraph 4.3.3.1.2.2 shall be examined to determine compliance with 3.4.3. Failure to comply shall be cause for rejection of the production represented by the sample.

4.3.3.5 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.3.5.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure of sample specimens to comply with requirements of the applicable drawing shall be cause for taking corrective action recommended by the applicable specification.

4.3.4.1 First article sample.

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4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

Table I

Phase	Quantity
Excess pressure	10
Service pressure	10

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

Table II

Phase	Quantity
Excess pressure	10
Service pressure	10

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II for this phase will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production represented by the production lot sample. An attempt will be made to recover all projectile fired in this phase.

4.3.4.4 Service pressure.-The samples provided in accordance with Tables I and II for this phase will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production list represented by the production lot sample.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists IEL-10524305 identifies the inspection equipment required to perform the examination and tests prescribed herein. Reference is made to the IEL for definition of coding format.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.-After final heat treatment or stress relief, but prior to finish machining, a Rockwell hardness test shall be performed on the outside surface of the projectile body to the rear of the location of the rear rotating band seat or at any other location deemed necessary for process control. The equipment and procedures for performing this test are prescribed by Federal Test Method Standard No. 151.

4.4.1.2 Tension test.

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4.4.1.2.1 Body.-From the samples provided by the heat treat or stress relief lots, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151, shall be taken longitudinally from opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seats. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be as prescribed by Federal Test Method Standard No. 151; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 Percent).

4.4.1.2.2 Rotating band.-The sample provided by 4.3.3.1.2 shall have the rotating band removed and carefully flattened without hammering so as to hold additional cold work to a minimum. The specimen blank shall be machined so that all serration marks from the band seat are removed. A tension test specimen having a one inch gage length and a one half inch gage width with grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by Federal Test Method Standard No. 151.

4.4.1.2.3 Base plug.-From the sample supplied in accordance with 4.3.3.1.2.5, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component or raw material coupon and tested in accordance with the procedures prescribed by Federal Test Method Standard No. 151.

4.4.2 Magnetic particle inspection.-The procedures for performing this test shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, both circular and longitudinal magnetization shall be used. The solid content of suspension shall be maintained at 0.2 to 0.4 milliliter (ML) of solids per 100 ML of suspension when oil is used, or 0.15 to 0.35 ML of solids per 100 ML of suspension when water is used. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. The magnetic particle test shall be

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4.4.2 (Cont'd)

performed on equipment that complies with the requirements of PA-PD-792. The operator(s) of the equipment and the inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.4.3 Hydrostatic test.-The body assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. The assembly shall be mounted in an approved test fixture in such a manner that it can be tested without constraint. In performing the test, the hydrostatic pressure will be applied gradually until the minimum required pressure is reached, afterwhich the pressure will be held for the minimum time required by the applicable drawing. The equipment shall be calibrated prior to the start of each day's operation and at intervals not over 4 hours continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of 2 hours or more.

4.4.4 Band seating.

4.4.4.1 Tightness.

4.4.4.1.1 Non-destructive method.-The sample provided by 4.3.3.4.1.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.3).

4.4.4.1.2 Destructive method.-The sample provided in accordance with 4.3.3.4.1.2 need not be processed through the band machining line, although the band should have a light surface cut to true the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on 2 points along the cylindrical portion of their length, near the front and rear edges. This comprises 6. diametrical measurements per band; these points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

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4.4.4.2 Sidewall contact.-The samples provided in accordance with 4.3.3.1.2.4 shall be sectioned by cutting through one plane normal to the longitudinal axis one-half inch forward of the forward rotating band. The cylindrical portion containing the rotating bands shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the two bands swaged into the band seats. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.4.3

4.4.5 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

4.4.5.1 Salt spray.-Procedures and equipment to be used in performing this test shall be as prescribed in Federal Test Method Standard No. 141.

4.4.6 Functional.

4.4.6.1 Excess pressure.-The projectiles supplied in accordance with 4.3.4, inert loaded and assembled into a complete round, will be fired for recovery in a gun for which the projectile is standard, the tube of which will have a minimum of 25 percent life remaining, based on current gun tube condemnation criteria. A propellant charge adjusted to produce a 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6.

4.4.6.2 Service pressure.-The projectile supplied in accordance with 4.3.4 inert loaded and assembled into a complete round will be fired alternately with projectiles from a reference lot for comparison. The projectile shall be fired in a gun for which the round is standard, the tube of which shall have a minimum of 25 percent life remaining based on current gun tube condemnation criteria. All rounds will be fired to impact on a vertical target placed approximately 2000 yards from the gun muzzle, accuracy in terms of maximum spread both horizontal and vertical will be recorded for evaluation by the technical agency. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6. Velocity and chamber pressure will be recorded for each round fired.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking, level C.-Packing and marking shall be in accordance with MIL-STD-1169 and this specification.

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5.2 Packing.

5.2.1 Body assembly.-Body assembly, part number 10524305, shall be packaged in accordance with Drawing F8837835.

5.2.2 Plug, base.-Thirty-two base plugs shall be packed, in accordance with MIL-STD-1169, in a 4 x 4 pattern, 1 to a cell, in 2 layers, using interlocking separators.

5.3 Marking.

5.3.1 Body assembly.-Palletized body assemblies shall be marked in accordance with Drawing F8837835.

5.3.2 Plug, base.-Packaged base plug will be marked in accordance with paragraph 3.4 of MIL-STD-1169.

5.4 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and divisions between lots clearly indicated to prevent mixing in transit.

5.5 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractor's plant in compliance with 3.5.3.

6.3 Band seating.-The operating procedures for use of Detroit Testing Machine as prepared by the Quality Branch, U. S. Army, Frankford Arsenal, SMUFA-Q3200.

6.4 The following materials complying with MIL-C-16173, Grade 2, have been found to be compatible with composition A-3.

a. Cosmoline No. 1102 - E. F. Haughton Co., Phila., Pa.

b. Nox-Rust No. 207, Grade 2 - Daubert Chemical Co., Chicago, Illinois.

c. Petrotect 2 - Pennsylvania Refining Co., Butler, Pa.

d. Rust Stop #1 - B & B Chemical Co., Miami, Florida.

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6.5 It is the responsibility of the supplier of the projectile metal parts, to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of forming and machining, to produce the shell to the dimensions and properties specified as an end item requirement.

6.6 The supplier of the selected steel raw material is responsible only for furnishing a metal composition grade and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations, identifiable with the standard AISI grade selected by the projectile manufacturer and complying with Specification ASTM, A576 special bar quality.

6.7 In the application of carbon steel, blooms, billets and slabs, forging quality processing techniques adapted to use of selected steel should be applied with the expectation that there will be variables in the metal cross sectional dimensions in weight per unit length in the hardenability variance from heat to heat, in the mechanical condition of the surface and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices and processing techniques having capabilities of compensating for such variables.

6.8 Heat treatment is understood to be any thermal process which brings about a modification in the mechanical or metallurgical properties of the metal involved.

6.9 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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